

真核有参转录组项目材料方法写作模板

1. RNA extraction

Total RNA was extracted from the tissue using TRIzol® Reagent according the manufacturer's instructions(Invitrogen) and genomic DNA was removed using DNase I (TaKara). Then RNA quality was determined using 2100 Bioanalyser (Agilent) and quantified using the ND-2000 (NanoDrop Technologies). High-quality RNA sample ($OD_{260/280}=1.8\sim2.2$, $OD_{260/230}\geq 2.0$, $RIN\geq 6.5$, $28S:18S\geq 1.0$, $>10\mu g$) is used to construct sequencing library.

2. Library preparation, and Illumina Hiseq sequencing

RNA-seq transcriptome libraries were prepared following TruSeq™ RNA sample preparation Kit from Illumina (San Diego, CA), using 1 μg of total RNA. Shortly, messenger RNA was isolated with polyA selection by oligo(dT) beads and fragmented using fragmentation buffer. cDNA synthesis, end repair, A-base addition and ligation of the Illumina-indexed adaptors were performed according to Illumina's protocol. Libraries were then size selected for cDNA target fragments of 200–300 bp on 2% Low Range Ultra Agarose followed by PCR amplified using Phusion DNA polymerase (NEB) for 15 PCR cycles. After quantified by TBS380, Paired-end libraries were sequenced by Illumina NovaSeq 6000 sequencing (150bp*2, Shanghai BIOZERON Co., Ltd).

3. Reads quality control and mapping

The raw paired end reads were trimmed and quality controlled by Trimmomatic with parameters (SLIDINGWINDOW:4:15 MINLEN:75) (version 0.36 <http://www.usadellab.org/cms/uploads/supplementary/Trimmomatic>). Then clean reads were separately aligned to reference genome with orientation mode using hisat2 (<https://ccb.jhu.edu/software/hisat2/index.shtml>) software. This software was used to map with default parameters. The quality assessment of these data were taken by qualimap_v2.2.1

(<http://qualimap.bioinfo.cipf.es/>) . Use htseq

(https://htseq.readthedocs.io/en/release_0.11.1/)to count each gene reads.

4. Differential expression analysis and Functional enrichment

To identify DEGs (differential expression genes) between the two different samples, the

expression level for each gene was calculated using the fragments per kilobase of exon per million mapped reads (FRKM) method. R statistical package edgeR (Empirical analysis of Digital Gene Expression in R, <http://www.bioconductor.org/packages/release/bioc/html/edgeR.html/>) was used for differential expression analysis. The DEGs between two samples were selected using the following criteria: the logarithmic of fold change was greater than 2 and the false discovery rate (FDR) should be less than 0.05. To understand the functions of the differentially expressed genes, GO functional enrichment and KEGG pathway analysis were carried out by Goatools (<https://github.com/tanghaibao/Goatools>) and KOBAS (<http://kobas.cbi.pku.edu.cn/home.do>). DEGs were significantly enriched in GO terms and metabolic pathways when their Bonferroni-corrected P-value was less than 0.05.

5. Alternative Splice events Identification

Alternative splice events that occurred in the samples were identified by Multivariate Analysis of Transcript Splicing (MATS, <http://rnaseq-mats.sourceforge.net/>). Only the isoforms that were similar to the reference or comprised novel splice junctions were considered, and the splicing differences were detected as exon inclusion, exclusion, alternative 5', 3', and intron retention events[5].

6. WGCNA construct

Briefly, WGCNA constructs networks using the absolute value of the Pearson's correlation coefficient as the gene co-expression measure, which is raised to a power to create the adjacency matrix. The topological overlap distance calculated from the adjacency matrix is then clustered with the average linkage hierarchical clustering. Our modules were defined using the cutreeDynamic function with a minimum module size of 30 genes. A module eigengene distance threshold of 0.25 was used to merge highly similar modules use mergeCloseModules function. We empirically set the cut-off of the weight value as 0.10 to determine biologically significant edges for each module. Correlation analyses between module eigengenes and measured agronomic traits were applied to explore the biological significances for each module. A gene with top 15% number of edges in a module was determined as the hub gene.